NOTES FOR ANALYSIS AND PDES

Based on Math 719 (Cole), Math 720 (M. Ifrim), Folland, etc.

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This is a checklist of incomplete proofs.

1. Proof that the Schwartz space is Fréchet in the appendix.

2. Proof for the negative Sobolev space is not finished yet.

3.Proof of smooth function approximation and partition by unity.

1 Introduction

Partial differential equations (PDEs) are equations involving the partial derivatives of functions of several variables. A general k-th order PDE for a function $u : \mathbb{R}^d \to \mathbb{R}^m$ has the form

$$F\big(D^k u(x), D^{k-1}u(x), \dots, Du(x), u(x), x\big) = 0,$$

where $D^\ell u$ denotes the collection of all partial derivatives of order ℓ.

PDEs were first introduced to model physical systems, and we will often use physical intuition to derive and understand them. However, they are not limited to mathematical physics: they also appear naturally in many parts of pure mathematics and in quantitative sciences.

In practice, one usually studies a PDE on a proper subset of $\mathbb{R}^d$ together with initial or boundary conditions. The main questions are existence, uniqueness, and qualitative behavior of solutions. Since the behavior depends strongly on the structure of the equation, it is usually impossible to build one theory that treats all PDEs at once. Instead, one organizes PDEs into several important classes and develops tools adapted to each class.

A first useful distinction is between linear and nonlinear equations. We say a PDE is linear if it has the form

$$a_k(x)D^k u + a_{k-1}(x)D^{k-1}u + \cdots + a_1(x)Du + a_0(x)u = f(x),$$

where the coefficients a_j may depend on x. Linear PDEs remain central for several reasons:

1. they are often the simplest models arising from physical considerations,
2. nonlinear equations frequently share broad features with their linearizations,
3. many nonlinear problems are first approached through linear theory.

In this set of notes, we will mainly focus on linear PDEs, with occasional nonlinear examples. Even among linear equations, the behavior varies greatly depending on the order and sign structure of the derivatives. It is therefore useful to organize the subject around several representative classes.

Class	Representative equation	Name
First-order	$\partial_t u + \operatorname{div}(uv) = 0$	Transport
Elliptic	$\Delta u = 0$	Laplace
Parabolic	$\partial_t u = \Delta u$	Heat
Hyperbolic	$\partial_t^2 u - \partial_x^2 u = 0$	Wave
Dispersive	$i\partial_t u + \Delta u = 0$	Schrödinger

These classes are not exhaustive, but they include many of the most important PDEs appearing in applications. In particular, the Laplace equation, the heat equation, and the wave equation will occupy a central role in what follows.

There are also important connections between these classes. Elliptic equations often appear as equilibrium versions of parabolic problems; transport and hyperbolic equations are often regularized by parabolic terms; and hyperbolic and dispersive equations share

several structural features. Thus, while classification is useful, one should also keep in mind the broader unity of the subject.

Part I

Sobolev Spaces

In the following chapters, we denote $\mathcal{D}(U)$ to be the test functions on an open set $U \subset \mathbb{R}^m$, i.e. the continuous functions with compact support on U, and V' to be the dual space of the vector space V.

2 Introduction to Sobolev Spaces

2.1 Sobolev spaces

Definiton 2.1.1. Let $u \in \mathcal{D}'(U)$, where $U \subset \mathbb{R}^d$ is open. The k-th order *L^p-based Sobolev norm* of u is

$$\|u\|_{W^{k,p}(U)} := \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p}, \quad 1 \leq p < \infty,$$

where we are using the distributional derivative and assume that $D^\alpha u$ is an L^p function.

Definiton 2.1.2. Let U be an open subset of $\mathbb{R}^d$. The *L^p-based Sobolev space* of order k on U is

$$W^{k,p}(U) = \{u \in D'(U) : \|u\|_{W^{k,p}(U)} < \infty\}.$$

Note that $\mathcal{D}(U) \subseteq W^{k,p}(U)$.

Definiton 2.1.3. The set of $u \in W^{k,p}(U)$ that vanish (to appropriate orders) on ∂U is

$$W_0^{k,p}(U) = \overline{C_c^\infty(U)}^{\|\cdot\|}_{W^{k,p}} \subset W^{k,p}(U).$$

When $p = 2$, we introduce the notation

$$H^k(U) = W^{k,2}(U), \qquad H_0^k(U) = W_0^{k,2}(U).$$

We can define an inner product on H^k by

$$\langle u, v \rangle_{H^k} = \sum_{|\alpha| \leq k} \langle D^\alpha u, D^\alpha v \rangle_{L^2}.$$

Proposition 2.1.1.

1. For all $k \in \mathbb{Z}_{\geq 0}$ and $1 \leq p \leq \infty$, the following Sobolev spaces $(W^{k,p}(U), \|\cdot\|_{W^{k,p}})$ and $(W_0^{k,p}(U), \|\cdot\|_{W^{k,p}})$ are Banach spaces.
2. For all $k \in \mathbb{Z}_{\geq 0}$, $(H^k(U), \langle \cdot, \cdot \rangle_{H^k})$ and $(H_0^k(U), \langle \cdot, \cdot \rangle_{H^k})$ are Hilbert spaces.
3. (Fourier-analytic characterization of H^k) Given $u \in H^k(U)$

$$\|u\|_{H^k} \simeq \|\hat{u}\|_{L^2} + \||\xi|^k \hat{u}\|_{L^2} \simeq \|(1 + |\xi|^2)^{k/2} \hat{u}\|_{L^2},$$

where $\simeq$ between norms are the equivalence.

Proof. □

2.2 Negative Sobolev Spaces

It is natural to consider the dual space of the Sobolev spaces, which can be interpreted as the following space as well.

Definiton 2.2.1. For $k > 0$ and $1 < p < \infty$, define the *negative order Sobolev norm*

$$\|u\|_{W^{-k,p}(U)} = \inf \left\{ \sum_{|\alpha| \le k} \|g_\alpha\|_{L^p} : u = \sum_{|\alpha| \le k} D^\alpha g_\alpha \right\}.$$

and the *negative order Sobolev space* is

$$W^{-k,p}(U) = \left\{ u \in \mathcal{D}'(U) : u = \sum_{|\alpha| \le k} D^\alpha g_\alpha,\ g_\alpha \in L^p(U) \right\}.$$

Proposition 2.2.1. For $k \in \mathbb{Z}_{\ge 0}$ and $1 < p < \infty$,

$$\left(W_0^{k,p}(U)\right)^* = W^{-k,q}(U), \qquad \frac{1}{p} + \frac{1}{q} = 1.$$

Proof. For $u \in \mathcal{D}(U)$ and some

$$v = \sum_{|\alpha| \le k} D^\alpha g_\alpha, \quad g_\alpha \in L^q(U),$$

and we may define $(\cdot)^* : W^{-k,q}(U) \to (W_0^{k,p}(U))^*$ by

$$v^\flat(u) = \int vu = \sum_{|\alpha| \le k} (-1)^{|\alpha|} \int g_\alpha D^\alpha u$$

which is well-defined since another representation is equal to this from the distributions' perspectives on $\mathcal{D}(U)$, where the equality can be extended to the whole $W_0^{k,p}(U)$ by the density argument that

$$|v^\flat(u)| \le \sum_{|\alpha| \le k} \left| \int g_\alpha u \right| \le \sum_{|\alpha| \le k} ||g_\alpha||_{L^q} ||D^\alpha u||_{L^p} \le 2||v||_{W^{-k,q}(U)} ||u||_{W^{k,p}(U)}$$

where we can choose the representation of v to make

$$\sum_{|\alpha| \le k} ||g_\alpha||_{L^p} \le 2||v||_{W^{-k,q}(U)}$$

and this help us to make the extension.

From the other side, we need to show for any linear functional on $W_0^{k,p}(U)$ can be obtained by some element in the negative Sobolev space. Firstly, notice there is a natural inclusion of $W_0^{k,p}(U)$ to $L^p(U)^{\oplus K(k)}$, and define a linear map

$$T : \mathcal{D}(U) \to L^p(U)^{\otimes K(k)}, \quad u \mapsto (u, D_{x_1} u, \cdots, D_{x^k} u, \cdots, D^\alpha u)$$

and then we may see that T is an isometry between

$$(\mathcal{D}(U), ||\cdot||_{W^{k,p}}) \text{ and } (T(\mathcal{D}(U)), |\cdot|_{L^p(U)^{\otimes K(k)}}).$$

For a linear functional l on $W_p^{k,p}(U)$, we may know l will induce a bounded linear functional

$$\tilde{l} : T(\mathcal{D}(U)) \to \mathbb{R}, \quad \tilde{l}(Tu) = l(u)$$

which can be extended to the whole $L^p(U)^{\otimes K(k)}$ by Hahn-Banach and denote the extended linear functional to be l'.

□

2.3 Relationship between a priori estimates, existence, and uniqueness

Proposition 2.3.1. Let X, Y be Banach spaces, and $P : X \to Y$ be bounded linear operator. Denote $P^* : Y^* \to X^*$ as the adjoint, i.e. $\langle v, Pu\rangle = \langle P^*v, u\rangle$. Assume for any $u \in X$ we have $\|u\|_X \leq C\|Pu\|_Y$, then

1. If $Pu = 0$, then $u = 0$.
2. For all $g \in X^*$, there exists $v \in Y^*$ such that

$$P^*v = g, \qquad \|v\|_{Y^*} \leq C\|g\|_{X^*}.$$

Proof. (1) is obvious. For (2), consider $PX := \{y \in Y, \exists\ x \in X \text{ s.t.} Px = y\}$ which is a subspace of Y, and define $\bar{v}(Px) = \langle g, x\rangle$ which is a well-defined linear functional on PX with

$$||\bar{v}(Px)|| \leq ||g||_{X^*}||x||_X \leq C||g||_{X^*}||Px||_Y$$

and hence we know there is a linear functional $v \in Y^*$ such that $v|_{PX} = \bar{v}$ and $||v||_{Y^*} \leq C||g||_{X^*}$ and we are done. □

Definiton 2.3.1. Let X be a Banach space, we call it *reflexive* if $u \to (v \mapsto \langle v, u\rangle)$ is an isomorphism.

Proposition 2.3.2. Assume X is reflexive. Let X, Y be Banach spaces, and let $P : X \to Y$ be bounded linear operator. Suppose $||v||_{Y^*} \leq C||P^*v||_{X^*}$. Then

1. If $v \in Y^*$ and $P^*v = 0$, then $v = 0$.
2. For all $f \in Y$, there exists a $u \in X$ such that $Pu = f$ and $||u||_X \leq C||f||_Y$.

Remark. All sobolev spaces $W_0^{k,p}(U)$ with $1 < p < \infty$ are reflexive.

Definiton 2.3.2. Given $U \subset Y$, the *annihilator* of U is

$$U^\perp := \{v \in Y^* : \langle v, f\rangle = 0, \text{ for any } f \in U\}$$

and similarly, for some $V \subset X^*$, the *annihilator* of V is

$${}^\perp V := \{u \in X : \langle g, u\rangle = 0, \text{ for any } g \in V\}$$

Proposition 2.3.3. For the operators in (2.3.1) we have

$$(\text{ran } P)^{\perp} = \ker(P^*), \quad \ker(P) =^{\perp} (\text{ran } P^*)$$

Proof. For the second equality, refer to Hahn-Banach to show that if some $u \in X$ is in the kernel of any linear functional in X^*, then $u = 0$. □

Proposition 2.3.4. Let X, Y be Banach spaces, with X reflexive, and $P : X \to Y$ be a bounded linear operator. If $P(X) = Y$, then thre exists some $C > 0$ such that

$$||v||_{Y^*} \leq C||P^* v||_{X^*}$$

Example 2.3.1. Try to show the 1-dimensional Laplace equation

$$\begin{cases} -u'' = f & \text{in } (0,1), \\ u = 0 & \text{at } x = 0, 1 \end{cases}$$

will have a unqiue solution in $H^1(0,1)$.

2.4 Approximation by smooth functions and smooth partition of unity

Lemma 2.4.1. Let ϕ be smooth, compactly supported i.e. $\phi \in C_0^\infty(U)$, and have

$$\int \phi dx = 1.$$

Let $u \in L^p(\mathbb{R}^d)$ with $1 \leq p < \infty$. Denote mollifiers

$$\phi_\epsilon(x) = \frac{1}{\epsilon^d}\phi(x/\epsilon) \quad (\text{so} \int \phi_\epsilon = 1).$$

Then

$$\|\phi_\epsilon * u - u\|_{L^p} \xrightarrow[\epsilon \to 0]{} 0,$$

where

$$\phi_\epsilon * u = \int \phi_\epsilon(x-y)u(y)dy.$$

Proof. □

Lemma 2.4.2. Suppose $\{U_\alpha\}_{\alpha \in A}$ be an open covering of U in $\mathbb{R}^d$. There exists a smooth partition of unity $\{\chi_\alpha\}_{\alpha \in A}$ on U subordinate to $\{U_\alpha\}_{\alpha \in A}$, i.e. 1. $\sum_\alpha \chi_\alpha(x) = 1$ on U and for all $x \in U$ there exist only finitely many nonzero $\chi_\alpha(x)$ 2. $\operatorname{supp} \chi_\alpha \subseteq U_\alpha$ 3. χ_α is smooth.

Proof. Start from a continuous partition of unity and apply the previous lemma to approximate by smooth functions. □

3 Approximation in Bounded Domains and the Extension Theorem

3.1 Approximation theorems in bounded domains

Theorem 3.1.1. Let $k \geq 0$ be an integer and $1 \leq p < \infty$. (i) $C^\infty(\mathbb{R}^d)$ is dense in $W^{k,p}(\mathbb{R}^d)$. (ii) $C_c^\infty(\mathbb{R}^d)$ is dense in $W^{k,p}(\mathbb{R}^d)$.

Proof. (a) This is an application of mollification. (b) Approximate by $f\chi(1/R)$, letting $R \to \infty$, where $\chi \in C_c^\infty(\mathbb{R}^d)$ is such that $\chi(0) = 1$. □

Theorem 3.1.2. Let $k \geq 0$ be an integer, $1 \leq p < \infty$, and U an open subset of $\mathbb{R}^d$. Then $C^\infty(U)$ is dense in $W^{k,p}(U)$.

Proof. □

3.2 The extension theorem

Theorem 3.2.1 (Extension theorem)**.** Let $k \geq 0$ be a nonnegative integer, $1 \leq p < \infty$, and U a bounded domain with C^k boundary. Let V be an open set such that $V \supseteq \bar{U}$. Then there exists an operator

$$\mathcal{E} : W^{k,p}(U) \to W^{k,p}(\mathbb{R}^d)$$

such that (i) (Extension) $\mathcal{E}u|_U = u$ (ii) (Linear and bounded) $\mathcal{E}$ is linear, and $\|\mathcal{E}u\|_{W^{k,p}(\mathbb{R}^d)} \leq C\|u\|_{W^{k,p}(U)}$. (iii) (Support prescription) $\operatorname{supp} \mathcal{E}u \subseteq V$.

Proof. □

4 Trace and Extension Theorems and Introduction to Sobolev Inequalities

4.1 The trace theorem

Definiton 4.1.1. For $u \in C^1(\overline{U})$, we define the trace to be

$$\mathrm{tr}_{\partial U} u := u|_{\partial U}.$$

We wish to extend this operation to all of $W^{1,p}(U)$. Note that $\mathrm{tr}_{\partial U}$ is linear, so we can extend it if we know it is bounded.

Theorem 4.1.1 (Trace theorem, non- sharp)**.** Let U be a bounded open subset of $\mathbb{R}^d$ with C^1 boundary ∂U , and let $1 < p < \infty$. Then for $u \in C^1(\overline{U})$

$$\|\mathrm{tr}_{\partial U} u\|_{L^p(\partial U)} \leq C\|u\|_{W^{1,p}(U)}.$$

(i) As a consequence, $\mathrm{tr}_{\partial U}$ extends uniquely by continuity (and density of $C^1(\overline{U}) \subset W^{1,p}(U)$) to a bounded linear map

$$\mathrm{tr}_{\partial U} : W^{1,p}(U) \to L^p(\partial U).$$

(ii) Moreover, $u \in W_0^{1,p}(U)$ if and only if $\mathrm{tr}_{\partial U} u = 0$.

Remark 4.1.2. The map $\mathrm{tr}_{\partial U} : W^{1,p}(U) \to L^p(\partial U)$ is not surjective.

Definiton 4.1.2. The *fractional (L^2-based) Sobolev spaces* is defined by the set of L^2 function u with

$$\|u\|_{H^k}^2 \simeq \|(1+|\xi|^2)^{s/2}\hat{u}(\xi)\|_{L^2_\xi}^{@} < \infty$$

which is also denoted as H^s.

Theorem 4.1.3 (Sharp trace theorem)**.** For $u \in C^\infty(\overline{\mathbb{R}^d_+}) \cap H^1(\mathbb{R}^d_+)$,

$$\|\,\mathrm{tr}_{\partial U}\, u\|_{H^{1/2}(\mathbb{R}^{d-1})} \leq C\|u\|_{H^1(\mathbb{R}^d_+)}.$$

Proof. □

Theorem 4.1.4. There exists a bounded linear map

$$\mathrm{ext}_{\partial U} : H^{1/2}(\mathbb{R}^{d-1}) \to H^1(\mathbb{R}^d_+)$$

such that

$$\mathrm{tr}_{\partial U} \circ \mathrm{ext}_{\partial U} = id.$$

Remark 4.1.5. This theorem asserts that every boundary function with the optimal trace regularity can be realized as the boundary value of an interior H^1 function.

Proof. We use the Poisson semigroup. Given $g \in \mathcal{S}(\mathbb{R}^{d-1})$, define its Poisson extension v by

$$v(x', x_d) = (P_{x_d} * g)(x'), \qquad \widehat{P_{x_d}}(\xi') = e^{-x_d|\xi'|}.$$

Equivalently, v is the unique solution to the Dirichlet problem

$$\begin{cases} \Delta v = 0 & \text{in } \mathbb{R}^d_+, \\ v(x', 0) = g(x') & \text{on } \partial\mathbb{R}^d_+. \end{cases}$$

Choose $\eta \in C_c^\infty(\mathbb{R})$ such that $\eta = 1$ for $|s| < 1$ and $\eta = 0$ for $|s| > 2$, and define

$$u = \mathrm{ext}_{\partial U}\, g := \eta(x_d)v.$$

Then

$$\mathcal{F}_{x'}u(\xi', x_d) = \eta(x_d)e^{-x_d|\xi'|}\hat{g}(\xi').$$

One checks that $u \in H^1(\mathbb{R}^d_+)$ and that

$$\|u\|_{H^1(\mathbb{R}^d_+)} \lesssim \|g\|_{H^{1/2}(\mathbb{R}^{d-1})}.$$

Moreover, since $\eta(0) = 1$, we have

$$\mathrm{tr}_{\partial U}\, u = g.$$

Therefore this construction gives a bounded right inverse to the trace map. □

Remark 4.1.6.
By a partition of unity and boundary straightening, one can define $H^{1/2}(\partial U)$ for C^1 boundaries. This also gives the sharp trace theorem in that setting.

Thus far we have described Sobolev regularity in terms of weak derivatives and L^p norms. There is another complementary point of view, especially natural in the L^2 setting, based on the Fourier transform. Roughly speaking, irregular behavior corresponds to large oscillation at small scales, and the Fourier transform is designed to isolate such oscillations.

For $f \in L^1(\mathbb{R}^d)$, define the Fourier transform by

$$\hat{f}(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix\cdot\xi} f(x)\, dx.$$

With the same normalization, the inverse Fourier transform is given by

$$\check{g}(x) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{ix\cdot\xi} g(\xi)\, d\xi.$$

This normalization is convenient in L^2, since Plancherel's theorem takes the form

$$\|\hat{f}\|_{L^2(\mathbb{R}^d)} = \|f\|_{L^2(\mathbb{R}^d)}.$$

In particular, the Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^d)$.

The key point for PDE is that differentiation becomes multiplication in frequency space. Formally, if f is smooth and decays sufficiently fast, then

$$\widehat{\partial_j f}(\xi) = i\xi_j \hat{f}(\xi),$$

and more generally,

$$\widehat{\partial^\alpha f}(\xi) = (i\xi)^\alpha \hat{f}(\xi).$$

Similarly, convolution becomes multiplication:

$$\widehat{f * g}(\xi) = (2\pi)^{d/2} \hat{f}(\xi)\hat{g}(\xi).$$

These identities explain why the Fourier transform is especially effective for linear PDE with constant coefficients.

Proposition 4.1.7 (Fourier characterization of H^k)**.** Let $k \in \mathbb{Z}_{\geq 0}$. Then for $f \in L^2(\mathbb{R}^d)$,

$$f \in H^k(\mathbb{R}^d) \qquad \Longleftrightarrow \qquad (1 + |\xi|^2)^{k/2} \hat{f}(\xi) \in L^2(\mathbb{R}^d).$$

Moreover,

$$\|f\|_{H^k}^2 \simeq \int_{\mathbb{R}^d} (1 + |\xi|^2)^k |\hat{f}(\xi)|^2\, d\xi.$$

Proof. By Plancherel and the identity $\widehat{\partial^\alpha f} = (i\xi)^\alpha \hat{f}$, we have

$$\|\partial^\alpha f\|_{L^2}^2 = \int_{\mathbb{R}^d} |\xi^\alpha|^2 |\hat{f}(\xi)|^2\, d\xi.$$

Summing over all $|\alpha| \leq k$ yields

$$\|f\|_{H^k}^2 = \sum_{|\alpha|\leq k} \|\partial^\alpha f\|_{L^2}^2 \simeq \int_{\mathbb{R}^d} (1 + |\xi|^2)^k |\hat{f}(\xi)|^2\, d\xi,$$

since $\sum_{|\alpha|\le k} |\xi^\alpha|^2$ is comparable to $(1+|\xi|^2)^k$. $\square$

Remark 4.1.8. This point of view extends naturally to fractional orders. For $s \ge 0$, one may define

$$H^s(\mathbb{R}^d) = \Big\{ f \in L^2(\mathbb{R}^d) : (1+|\xi|^2)^{s/2} \hat{f} \in L^2(\mathbb{R}^d) \Big\},$$

with norm

$$\|f\|_{H^s}^2 = \int_{\mathbb{R}^d} (1+|\xi|^2)^s |\hat{f}(\xi)|^2 \, d\xi.$$

This also leads to negative Sobolev spaces and gives a clean frequency-space language for elliptic regularity.

Remark 4.1.9 (Poisson equation on the torus)**.** Suppose f is defined on $\mathbb{T}^d$ and we want to solve

$$\Delta u = f.$$

Taking Fourier series gives

$$-|k|^2 \hat{u}(k) = \hat{f}(k), \qquad k \in \mathbb{Z}^d.$$

Hence a necessary condition for solvability is $\hat{f}(0) = 0$, i.e. $\int_{\mathbb{T}^d} f = 0$. If this holds, then for $k \ne 0$ we may set

$$\hat{u}(k) = -|k|^{-2} \hat{f}(k),$$

while $\hat{u}(0)$ remains free, reflecting the fact that Δ has constants in its kernel. In frequency space one sees immediately that inverting Δ gains two derivatives, which is the basic elliptic regularity phenomenon.

5 Sobolev Inequalities

5.1 The Gagliardo–Nirenberg–Sobolev inequality

Theorem 5.1.1 (Gagliardo–Nirenberg–Sobolev inequality)**.** Let $d \ge 2$. For $u \in C_c^1(\mathbb{R}^d)$, we have

$$\|u\|_{L^{\frac{d}{d-1}}(\mathbb{R}^d)} \le \|Du\|_{L^1(\mathbb{R}^d)}.$$

Lemma 5.1.2 (Loomis–Whitney inequality)**.** Let $d \ge 2$. For $j = 1, \dots, d$, suppose

$$f_j = f_j(x_1, \dots, \widehat{x_j}, \dots, x_d).$$

Then

$$\left\| \prod_{j=1}^d f_j \right\|_{L^1(\mathbb{R}^d)} \le \prod_{j=1}^d \|f_j\|_{L^{d-1}(\mathbb{R}^{d-1})}.$$

Remark 5.1.3. The Loomis–Whitney inequality gives a geometric control of the measure of a set in $\mathbb{R}^d$ by the measures of its coordinate projections. In particular, if $E \subset \mathbb{R}^d$ and $\pi_j(E)$ denotes the j-th coordinate projection, then

$$|E| \le \prod_{j=1}^d |\pi_j(E)|^{1/(d-1)}.$$

Remark 5.1.4. The Gagliardo–Nirenberg–Sobolev inequality is the functional counterpart of the isoperimetric inequality. Via the layer-cake decomposition, one may reduce the functional statement to geometric information on level sets.

5.2 L^p-based Sobolev inequalities for $1 \le p < d$

Theorem 5.2.1 (Sobolev inequality for $1 \le p < d$)**.** Let $d \ge 2$ and assume $1 \le p < d$. For $u \in C_c^1(\mathbb{R}^d)$, we have

$$\|u\|_{L^q(\mathbb{R}^d)} \le C \|Du\|_{L^p(\mathbb{R}^d)}, \qquad q = \frac{dp}{d-p}.$$

Remark 5.2.2. The exponent $q = \frac{dp}{d-p}$ is dictated by scaling (or dimensional analysis):

$$[x]^{d/q} = [x]^{-1+d/p}.$$

The condition $p < d$ is therefore unavoidable in this regime.

Theorem 5.2.3. Let $d \ge 2$ and assume that $1 \le p < d$.

1. For $u \in W^{1,p}(\mathbb{R}^d)$,

$$\|u\|_{L^q(\mathbb{R}^d)} \le C \|Du\|_{L^p(\mathbb{R}^d)}, \qquad q = \frac{dp}{d-p}.$$

2. Let U be a bounded domain. For $u \in W_0^{1,p}(U)$,

$$\|u\|_{L^q(U)} \le C \|Du\|_{L^p(U)}, \qquad q = \frac{dp}{d-p}.$$

3. Let U be a bounded domain with C^1 boundary. For $u \in W^{1,p}(U)$,

$$\|u\|_{L^q(U)} \le C \|u\|_{W^{1,p}(U)}, \qquad q = \frac{dp}{d-p}.$$

Remark 5.2.4. In part (iii), one needs both $\|u\|_{L^p}$ and $\|Du\|_{L^p}$ in order to perform the extension argument. By contrast, in part (ii) the vanishing trace condition removes the need for the L^p term. This is why part (ii) is often referred to as a Poincaré or Friedrichs inequality.

5.3 L^p-based Sobolev inequalities for $p > d$

Lemma 5.3.1. Let $p > d$, $d \ge 2$, and $u \in C^\infty(\mathbb{R}^d)$. Then

$$\frac{1}{|B_r(x)|} \int_{B_r(x)} |u(x) - u(y)|\, dy \le C \int_{B_r(x)} \frac{|Du(y)|}{|x-y|^{d-1}}\, dy.$$

Theorem 5.3.2 (Morrey's inequality on $\mathbb{R}^d$)**.** Let $p > d$ with $d \ge 2$, and let $u \in C^\infty(\mathbb{R}^d)$. Then

$$|u(x) - u(y)| \le C|x-y|^\alpha \|Du\|_{L^p(\mathbb{R}^d)}, \qquad \alpha = 1 - \frac{d}{p}.$$

Remark 5.3.3. The exponent

$$\alpha = 1 - \frac{d}{p}$$

is again determined by scaling:

$$1 = \alpha + (-1) + \frac{d}{p}.$$

In particular, Morrey's inequality is scale invariant.

Definiton 5.3.1. Let $u \in C(U)$. The Hölder seminorm of order α is

$$[u]_{C^{0,\alpha}(U)} = \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x - y|^{\alpha}}.$$

Remark 5.3.4. This is only a seminorm: if $[u]_{C^{0,\alpha}(U)} = 0$, then u is constant rather than identically zero.

Definiton 5.3.2. The Hölder norm of order α is

$$\|u\|_{C^{0,\alpha}(U)} = [u]_{C^{0,\alpha}(U)} + \|u\|_{L^1(U)}.$$

The corresponding Hölder space is

$$C^{0,\alpha}(U) = \{u \in C(U) : \|u\|_{C^{0,\alpha}(U)} < \infty\}.$$

Theorem 5.3.5 (Morrey's inequality on bounded domains)**.** Let $d \geq 2$, let $p > d$, and let $U \subset \mathbb{R}^d$ be a bounded domain with C^1 boundary. If $u \in W^{1,p}(U)$, then $u \in C^{\alpha}(U)$ with

$$\alpha = 1 - \frac{d}{p}.$$

Moreover,

$$\|u\|_{C^{\alpha}(U)} \leq C \|u\|_{W^{1,p}(U)}.$$

5.4 The critical case $p = d$

Remark 5.4.1. The borderline case $p = d$ does *not* satisfy an estimate of the form

$$\|u\|_{L^{\infty}(U)} \leq C \|u\|_{W^{1,d}(U)}.$$

Thus the natural target space at the endpoint is not L^{∞}, but rather a slightly larger space.

Definiton 5.4.1. A function $u \in C^1$ has bounded mean oscillation (BMO) if

$$\|u\|_{\mathrm{BMO}} = \sup_{x \in \mathbb{R}^d,\, r > 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} \left| u(y) - \frac{1}{|B_r(x)|} \int_{B_r(x)} u \, dy \right| dy.$$

Example 5.4.1. Let $U = B_1(0) \subset \mathbb{R}^2$ and define

$$u(x) = \log \log \left(10 + \frac{1}{|x|} \right).$$

Then as $r = |x| \to 0$,

$$u(r) \sim \log\log\left(\frac{1}{r}\right) \to +\infty,$$

so u is unbounded in B_1 and does not extend continuously to the origin.

On the other hand, one computes

$$|\nabla u(x)| \sim \frac{1}{r\log(1/r)} \qquad (r \to 0),$$

and hence

$$\int_{B_1} |\nabla u|^p \, dx < \infty \quad \Longleftrightarrow \quad p \le 2.$$

Therefore, in dimension $d = 2$, we obtain an example with

$$u \in W^{1,2}(B_1), \qquad u \notin L^\infty(B_1), \qquad u \text{ is not continuous at } 0.$$

This shows that $W^{1,d}$ does not embed into L^∞ at the critical exponent.

Theorem 5.4.2 (Endpoint Sobolev–BMO embedding)**.** Let $d \ge 2$ and let $u \in C_c^\infty(\mathbb{R}^d)$. Then

$$\|u\|_{\mathrm{BMO}} \le C(d) \, \|\nabla u\|_{L^d(\mathbb{R}^d)} \, .$$

Remark 5.4.3. A key tool in the proof is the Hardy–Littlewood maximal function

$$Mu(x) = \sup_{r>0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |u|,$$

which satisfies the estimate

$$\|Mu\|_{L^p} \le C \, \|u\|_{L^p} \, , \qquad 1 < p \le \infty.$$

5.5 Compactness of Sobolev embeddings: general notions

Definiton 5.5.1. Let X, Y be normed spaces and let $T : X \to Y$ be linear. We say that T is a compact operator if $T(B_X)$, the image of the unit ball in X, is compact in Y. Equivalently, for every bounded sequence $\{x_n\} \subset X$, the sequence $\{Tx_n\}$ admits a convergent subsequence in Y.

Definiton 5.5.2. Suppose $\iota : X \to Y$ is an embedding, i.e. a bounded linear injective map. We say that the embedding $X \hookrightarrow Y$ is compact if ι is a compact operator.

Remark 5.5.1. For Sobolev spaces, one is interested in statements of the form

$$W^{1,p}(U) \Subset L^q(U),$$

where $\Subset$ denotes compact embedding. Compactness is substantially stronger than mere continuity of the embedding and is essential in many PDE arguments for obtaining strong convergence.

Remark 5.5.2. The basic compactness principle in this direction is the Arzelà–Ascoli theorem. It will later be combined with regularity estimates such as Morrey's inequality.

Part II

Linear Elliptic Equations

6 Basic definitions and examples

6.1 Elliptic operators and principal symbols

Elliptic PDEs may be viewed as a natural generalization of the Laplace equation

$$-\Delta u = f.$$

A key feature of the Laplacian is that its principal symbol is nonvanishing for every nonzero frequency, and this motivates the general notion of ellipticity.

Definiton 6.1.1. Let P be a partial differential operator. The *symbol* of P is obtained by replacing each derivative ∂_j by $i\xi_j$.

For instance,

$$-\Delta = -\sum_{j=1}^{d} \partial_j \partial_j$$

has principal symbol

$$-\sum_{j=1}^{d} (i\xi_j)^2 = |\xi|^2,$$

which is invertible for every $\xi \neq 0$.

Suppose now that P is a linear partial differential operator of order K acting on vector-valued functions $u = (u_J)_{J=1}^N : U \to \mathbb{R}^N$, with leading part of the form

$$(Pu)_I = \sum_{J,\, |\alpha|=K} A_{IJ,\alpha}(x) \partial^\alpha u_J + (\text{lower-order terms}).$$

Definiton 6.1.2. The *principal symbol* of P is

$$\sigma_{\text{prin}}(P)(x,\xi) = i^K \sum_{|\alpha|=K} A_{IJ,\alpha}(x) \xi^\alpha.$$

We say that P is *elliptic* if $\sigma_{\text{prin}}(P)(x,\xi)$ is invertible for every $x \in U$ and every $\xi \neq 0$.

In the scalar case $N = 1$, an operator of order K has the form

$$Pu = \sum_{|\alpha|=K} a_\alpha(x) \partial^\alpha u,$$

and then

$$\sigma_{\text{prin}}(P)(x,\xi) = i^K \sum_{|\alpha|=K} a_\alpha(x) \xi^\alpha.$$

The first genuinely interesting case is the second-order scalar operator

$$Pu = a_{ij}(x) \partial_i \partial_j u + b_i(x) \partial_i u + c(x) u.$$

Its ellipticity condition is

$$a_{ij}(x)\xi_i\xi_j \neq 0 \qquad \text{for all } x \in U,\ \xi \neq 0.$$

When $a = [a_{ij}]$ is symmetric, this is equivalent to positive definiteness of the matrix $a(x)$.

Definiton 6.1.3 (Uniform ellipticity)**.** A second-order operator with symmetric coefficient matrix $a = [a_{ij}]$ is *uniformly elliptic* if there exists $\lambda > 0$ such that

$$a_{ij}(x)\xi_i\xi_j \geq \lambda|\xi|^2 \qquad \text{for all } x \in U,\ \xi \in \mathbb{R}^d.$$

Equivalently, every eigenvalue of $a(x)$ is bounded below by λ uniformly in x.

Remark 6.1.1. Elliptic equations arise naturally in variational problems, in equilibrium models, and as part of evolutionary systems. They also exhibit strong regularity properties: roughly speaking, if $Pu = f$ and P is elliptic of order K, then one expects u to be more regular than f by K derivatives.

Example 6.1.1 (Pressure equation for incompressible Euler)**.** Let $u : \mathbb{R}_t \times \mathbb{R}^3 \to \mathbb{R}^3$ satisfy the incompressible Euler equations

$$\begin{cases} \partial_t u + u \cdot \nabla u + \nabla p = 0, \\ \nabla \cdot u = 0. \end{cases}$$

Taking divergence of the first equation and using $\nabla \cdot u = 0$ leads to an elliptic equation for the pressure,

$$-\Delta p = \nabla \cdot (u \cdot \nabla u).$$

Thus an elliptic problem appears naturally inside a fluid evolution equation.

6.2 Boundary value problems in the scalar case

From now on, we mainly consider the scalar case $N = 1$, assume $d \geq 2$, and work on a bounded domain $U \subset \mathbb{R}^d$ with sufficiently regular boundary. For second-order elliptic equations, the most important boundary value problem is the Dirichlet problem

$$\begin{cases} Pu = f & \text{in } U, \\ u = g & \text{on } \partial U. \end{cases}$$

Here the boundary datum is interpreted through the trace map when $u \in H^1(U)$.

A standard reduction shows that it is enough to study the homogeneous boundary condition $u|_{\partial U} = 0$. Indeed, if $\widetilde{g}$ is any extension of the boundary datum g with the appropriate regularity, then setting $v = u - \widetilde{g}$ reduces the problem to

$$\begin{cases} Pv = \widetilde{f} & \text{in } U, \\ v = 0 & \text{on } \partial U. \end{cases}$$

Thus the study of the general Dirichlet problem reduces to the zero-boundary case.

6.3 Divergence form operators

Definiton 6.3.1 (Divergence form operator)**.** Let $U \subset \mathbb{R}^d$ be open. A second-order operator is said to be in *divergence form* if

$$Pu = -\partial_i(a_{ij}\partial_j u) + \partial_i(b_i u) + cu,$$

where a_{ij}, b_i, and c are coefficient functions.

Remark 6.3.1. The minus sign in front of the principal part is chosen so that, under ellipticity assumptions, the operator is coercive at the level of energy estimates. This convention is standard in elliptic theory and is especially convenient for the Dirichlet problem.

If the coefficients are smooth enough, one may expand the operator into non-divergence form:

$$Pu = -a_{ij}\partial_i\partial_j u + \big(b_i - \partial_j a_{ij}\big)\partial_i u + (\partial_i b_i + c)u.$$

7 A priori estimates and solvability

7.1 An H^1 a priori estimate

The basic tool in existence and uniqueness theory is an a priori estimate.

Theorem 7.1.1 (A priori estimate)**.** Suppose $u \in H_0^1(U)$ solves the Dirichlet problem

$$\begin{cases} Pu = f & \text{in } U, \\ u = 0 & \text{on } \partial U, \end{cases}$$

where

$$Pu = -\partial_i(a_{ij}\partial_j u) + \partial_i(b_i u) + cu.$$

Assume that P is uniformly elliptic, that a_{ij}, b_i, c are smooth, and that $b, c \in L^\infty(U)$ with

$$\|b\|_{L^\infty(U)} + \|c\|_{L^\infty(U)} \le A.$$

Then there exist constants $C > 0$ and $\gamma \ge 0$ such that

$$\|u\|_{H^1(U)} \le C\,\|f\|_{H^{-1}(U)} + \gamma\,\|u\|_{L^2(U)}\,.$$

This estimate shows that, up to a lower-order L^2 term, the H^1 norm of a solution is controlled by the size of the forcing term in H^{-1}.

7.2 Mapping property and the adjoint operator

Under the assumptions $a_{ij}, b_i, c \in L^\infty(U)$, the operator P defines a bounded linear map

$$P : H^1(U) \to H^{-1}(U).$$

Indeed, the principal part is the distributional derivative of an L^2 function, while the lower-order terms lie in $L^2(U)$ and hence also define elements of $H^{-1}(U)$.

When the Dirichlet condition is imposed, one naturally restricts to

$$P : H_0^1(U) \to H^{-1}(U).$$

The corresponding adjoint operator with respect to the L^2 pairing is

$$P^* u = -\partial_j(a_{jk}\partial_k u) - \partial_j(b_j u) + cu,$$

assuming real-valued coefficients. In particular, the same kind of energy estimate applies to P^*.

7.3 A functional-analytic solvability lemma

The abstract principle underlying the solvability theory is the following.

Lemma 7.3.1. Let X and Y be Banach spaces, and let $P : X \to Y$ be bounded and linear.

1. If
$$\|u\|_X \le C \|Pu\|_Y ,$$
then $\ker P = \{0\}$, and for every $g \in X^*$ there exists $v \in Y^*$ such that
$$P^* v = g, \qquad \|v\|_{Y^*} \le C \|g\|_{X^*} .$$

2. If
$$\|v\|_{Y^*} \le C' \|P^* v\|_{X^*} ,$$
then $\ker P^* = \{0\}$, and for every $f \in Y$ there exists $u \in X$ such that
$$Pu = f, \qquad \|u\|_X \le C' \|f\|_Y .$$

In the elliptic setting one applies this with

$$X = H_0^1(U), \qquad Y = H^{-1}(U).$$

7.4 The purely divergence-form case

When the lower-order terms vanish, the L^2 term in the a priori estimate can be removed.

Lemma 7.4.1. If $b = 0$ and $c = 0$, so that

$$Pu = -\partial_j(a_{jk}\partial_k u),$$

then the energy estimate holds with $\gamma = 0$, namely

$$\|u\|_{H^1(U)} \le C \|f\|_{H^{-1}(U)}$$

for solutions $u \in H_0^1(U)$ of $Pu = f$.

Theorem 7.4.2. For every $f \in H^{-1}(U)$, there exists a unique $u \in H_0^1(U)$ such that

$$-\partial_j(a_{jk}\partial_k u) = f \qquad \text{in } U.$$

Thus the homogeneous Dirichlet problem is uniquely solvable in the pure divergence-form case.

7.5 Fredholm alternative for general second-order elliptic operators

For general lower-order terms, the previous coercive argument is replaced by a Fredholm argument. The key idea is that, after adding a sufficiently large positive multiple of the identity, one obtains an invertible operator, and the remaining part becomes a compact perturbation of the identity.

Lemma 7.5.1 (Fredholm alternative for compact perturbations)**.** Let $K : X \to X$ be compact and set $T = I + K$.

1. $\ker(I + K)$ is finite dimensional.
2. There exists $n_0 \geq 1$ such that
$$\ker(I + K)^n = \ker(I + K)^{n_0} \qquad \text{for all } n \geq n_0.$$
3. $\operatorname{ran}\,(I + K)$ is closed and satisfies
$$\operatorname{ran}\,(I + K) = \big(\ker(I + K^*)\big)^{\perp}.$$
4. The dimensions of the kernels agree:
$$\dim \ker(I + K) = \dim \ker(I + K^*).$$

Applying this principle to the Dirichlet problem yields the following conclusion.

Theorem 7.5.2 (Fredholm alternative for elliptic Dirichlet problems)**.** Let P be a second-order scalar elliptic operator on a bounded domain U with C^1 boundary. Then exactly one of the following alternatives holds:

1. For every $f \in H^{-1}(U)$, there exists a unique $u \in H_0^1(U)$ such that
$$Pu = f,$$
and moreover
$$\|u\|_{H^1(U)} \leq C \|f\|_{H^{-1}(U)}.$$
2. There exists a nonzero $u \in H_0^1(U)$ such that
$$Pu = 0.$$

If the second alternative holds, then both $\ker P$ and $\ker P^*$ are finite dimensional, and the equation
$$Pu = f$$
is solvable if and only if
$$\langle f, v\rangle = 0 \qquad \text{for all } v \in \ker P^*.$$

Remark 7.5.3. This theorem shows that a second-order elliptic Dirichlet problem behaves, up to compact errors, like a finite-dimensional linear system: either one has uniqueness and full solvability, or else the only obstruction to solvability is orthogonality to the adjoint kernel.

8 L^2-based elliptic regularity

8.1 Interior regularity for the Poisson equation

A basic principle in elliptic theory is that solutions are more regular than the forcing term. For the Poisson equation

$$-\Delta u = f,$$

one expects a gain of two derivatives.

Definiton 8.1.1. Let $U \subset \mathbb{R}^d$ be open, let $V \Subset U$, and suppose u solves an elliptic equation on U. We say that *interior regularity* holds if one can estimate higher Sobolev norms of u on V in terms of the data on U.

For the Poisson equation, the model estimate is the following: if $f \in H^k(U)$, then one expects

$$\|u\|_{H^{k+2}(V)} \leq C \|f\|_{H^k(U)} + C \|u\|_{L^2(U)}.$$

Similarly, in Hölder spaces one expects estimates of the form

$$\|u\|_{C^{k+2,\alpha}(V)} \leq C \|f\|_{C^{k,\alpha}(U)} + C \|u\|_{L^\infty(U)}.$$

The first type is called L^2-based regularity theory, while the second belongs to Schauder theory.

Remark 8.1.1. The key idea in L^2-based regularity is the energy method: one multiplies the equation by a cutoff function times a suitable derivative of u, integrates by parts, and estimates the resulting terms. Because cutoff functions are used, these arguments are intrinsically interior.

Theorem 8.1.2 (Interior H^1 estimate for Poisson)**.** Let $V \Subset U$, and suppose that u solves

$$-\Delta u = f \qquad \text{in } U,$$

with $f \in L^2(U)$. Then

$$\|Du\|_{L^2(V)} \leq C \|f\|_{L^2(U)} + C \|u\|_{L^2(U)}.$$

Theorem 8.1.3 (Interior H^2 estimate for Poisson)**.** Let $V \Subset W \Subset U$, and suppose that u solves

$$-\Delta u = f \qquad \text{in } U,$$

with $f \in L^2(U)$. Then

$$\|u\|_{H^2(V)} \leq C \|f\|_{L^2(U)} + C \|u\|_{L^2(U)}.$$

Remark 8.1.4. The repeated use of nested domains $V \Subset W \Subset U$ is not accidental. Each cutoff function is chosen to be identically 1 on the smaller set and supported inside the larger one, so that integration by parts produces no boundary terms. When derivatives fall on the cutoff, one needs extra room between the region where norms are measured and the boundary of the ambient domain.

8.2 Interior H^2 regularity for general elliptic operators

Consider the divergence-form elliptic operator

$$Pu = -\partial_j(a_{jk}\partial_k u) + b_j\partial_j u + cu,$$

where $a(x) \geq \lambda I$ for some $\lambda > 0$ and where the coefficients are bounded. To obtain H^2 control, one also assumes enough regularity on the coefficients, typically $\partial a \in L^\infty(U)$.

Theorem 8.2.1 (H^2 elliptic regularity)**.** Let $u \in H^1(U)$ be a weak solution of

$$Pu = f \qquad \text{in } U,$$

with $f \in L^2(U)$. Assume that P is uniformly elliptic, that $a, b, c \in L^\infty(U)$, and that the coefficients have enough regularity to justify commuting with derivatives. Then for every $V \Subset U$,

$$u \in H^2(V) \qquad \text{and} \qquad \|u\|_{H^2(V)} \leq C\big(\|f\|_{L^2(U)} + \|u\|_{L^2(U)}\big).$$

Remark 8.2.2. At the formal level, the proof follows the same pattern as for the Poisson equation. However, because one does not yet know that u has second derivatives, one commutes the equation with difference quotients rather than with classical derivatives.

Definiton 8.2.1. Fix $k \in \{1, \dots, d\}$ and $h \in \mathbb{R} \setminus \{0\}$. The *difference quotient* in the e_k direction is defined by

$$D_k^h v(x) = \frac{v(x + he_k) - v(x)}{h}.$$

As $h \to 0$, this converges formally to $\partial_k v$.

Lemma 8.2.3. Let $V \Subset U$.

1. If $u \in W^{1,p}(U)$, then for $|h| \ll 1$,

$$\left\|D_k^h u\right\|_{L^p(V)} \leq C \left\|\partial_k u\right\|_{L^p(U)}.$$

2. Conversely, if $u \in L^p(U)$ and the difference quotients satisfy a uniform bound

$$\left\|D_k^h u\right\|_{L^p(V)} \leq A \qquad \text{for all sufficiently small } h,$$

then $\partial_k u \in L^p(V)$ and

$$\|\partial_k u\|_{L^p(V)} \leq A.$$

These facts allow one to repeat the energy estimate at the level of difference quotients and then pass to the limit.

8.3 Boundary H^2 regularity

Interior estimates alone do not control second derivatives up to the boundary. For boundary regularity one must combine the elliptic equation with the boundary condition and the geometry of ∂U.

Theorem 8.3.1 (Boundary H^2 regularity)**.** Let

$$Lu = -\partial_i(a_{ij}\partial_j u) + b_i\partial_i u + cu = f \qquad \text{in } U,$$

where P is uniformly elliptic and the coefficients satisfy the same assumptions as in the interior H^2 theorem. Assume in addition that

1. $u \in H_0^1(U)$,
2. ∂U is of class C^2.

Then

$$u \in H^2(U) \qquad \text{and} \qquad \|u\|_{H^2(U)} \leq C\big(\|f\|_{L^2(U)} + \|u\|_{L^2(U)}\big).$$

Remark 8.3.2. Near a boundary point, one first straightens the boundary by a C^2 change of coordinates, reducing the problem locally to a half-ball. Tangential second derivatives are then estimated by the interior argument, using that the Dirichlet condition forces tangential derivatives to vanish on the flattened boundary. The remaining normal second derivative is recovered from the equation itself, together with uniform ellipticity, which guarantees control of the coefficient of $\partial_d^2 u$.

A Topological Space

A.1 Radon Measure

Proposition A.1.1. If μ is a Radon measure on X, $C_c(X)$ is dense in $L^p(\mu)$ for $1 \leq p < \infty$.

B Functional Analysis

B.1 Banach Spaces

Theorem B.1.1 (Hahn-Banach Theorem)**.**
Let $\mathcal{X}$ be a real vector space, p a sublinear functional, i.e.

$$p(x+y) \leq p(x) + p(y), \quad p(\lambda x) = \lambda p(x) \text{ for any } x, y \in \mathcal{X} \text{ and } \lambda \geq 0$$

on $\mathcal{X}$, $\mathcal{M}$ a subspace of $\mathcal{X}$, and f a linear functional on $\mathcal{M}$ such that $f(x) \leq p(x)$ for all $x \in \mathcal{M}$. Then there exists a linear functional F on $\mathcal{X}$ such that $F(x) \leq p(x)$ for all $x \in \mathcal{X}$ and $F|_{\mathcal{M}} = f$.

Theorem B.1.2. Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measure spaces, and let K be an $(\mathcal{M} \otimes \mathcal{N})$-measurable function on $X \times Y$. Suppose that there exists $C > 0$ such that $\int |K(x,y)| d\mu(x) \leq C$ for a.e. $y \in Y$ and $\int |K(x,y)| d\nu(y) \leq C$ for a.e. $x \in X$, and that $1 \leq p \leq \infty$. If $f \in L^p(\nu)$, then the integral

$$Tf(x) = \int K(x,y) f(y) d\nu(y)$$

converges absolutely for a.e. $x \in X$ and $||Tf||_p \leq C||f||_p$.

Theorem B.1.3 (Minkowski Integral Inequality)**.** Suppose that $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ-finite measure spaces. Let f be an $(\mathcal{M} \otimes \mathcal{N})$-measurable function on $X \times Y$.

1. If $f \geq 0$ and $1 \leq p < \infty$, then

$$\left[\int \left(\int f(x,y) d\nu(y)\right)^p d\mu(x)\right]^{1/p} \leq \int \left[\int f(x,y)^p d\mu(x)\right]^{1/p} d\nu(y)$$

2. If $1 \leq p \leq \infty, f(\cdot, y) \in L^p(\mu)$ for a.e. y, and the function $y \mapsto ||f(\cdot, y)||_p$ is in $L^1(\nu)$, then $f(x, \cdot) \in L^1(\nu)$ for a.e. x and the function $x \mapsto \int f(x,y) d\nu(y)$ is in $L^p(\mu)$, and

$$\left\| \int f(\cdot, y) d\nu(y) \right\|_p \leq \int \|f(\cdot, y)\|_p d\nu(y)$$

B.2 Fourier Analysis

Definiton B.2.1. C_c^∞ denotes the space of smooth functions with compact support, where

$$\psi(x) = \begin{cases} \exp\left(\frac{1}{|x|^2 - 1}\right) & \text{if } |x| < 1, \\ 0 & \text{if } |x| \geq 1, \end{cases}$$

The *Schwartz space* $\mathcal{S}$ is defined by

$$\mathcal{S} := \{f \in C^\infty : ||f||_{(N,\alpha)} < \infty \text{ for all } N, \alpha\}$$

where

$$||f||_{(N,\alpha)} = \sup_{x\in\mathbb{R}^n} (1+|x|)^N |\partial^\alpha f(x)|$$

Definiton B.2.2 (Fréchet Space)**.** A *Fréchet space* is a complete Hausdorff topological vector space with a topology defined by a countable family of seminorms.

Proposition B.2.1. $\mathcal{S}$ is a Fréchet space with the topology defined by the norms $||\cdot||_{(N,\alpha)}$.

Proof. □

Proposition B.2.2. If $f \in C^\infty$, then $f \in \mathcal{S}$ iff $x^\beta \partial^\alpha f$ is bounded for all multi-indices α, β, iff $\partial^\alpha(x^\beta f)$ is bounded for all multi-indices α, β.

Proof. (1) $\implies$ (2) $\Leftrightarrow$ (3) is quite simple. For (2) $\implies$ (1), here is a good way to show it: for any integer N, we know that $\sum_{i=1}^n |x_i|^N$ has a positive minimum δ on the unit sphere. So we know that $\sum_{i=1}^n |x_i|^N \geq \delta |x|^N$, and then

$$(1+|x|)^N \leq 2^N(1+|x|^N) \leq 2^N(1+\delta^{-1}\sum_{i=1}^n |x_j^N|) \leq 2^N\delta^{-1}\sum_{|\beta|\leq N} |x^\beta|$$

□

Definiton B.2.3 (Shift Operator)**.**

If f is a function on $\mathbb{R}^m$ and $y \in \mathbb{R}^m$,

$$\tau_y f(x) = f(x-y)$$

f is called uniformly continuous if $||\tau_y f - f||_u \to 0$ as y.

Lemma B.2.3. If $f \in C_c(\mathbb{R}^n)$, then f is uniformly continuous.

Proposition B.2.4. If $1 \leq p < \infty$, translation is continuous in the L^p norm, i.e. if $f \in L^p$ and $z \in \mathbb{R}^n$, then $\lim_{y\to 0} ||\tau_{y+z} f - \tau_z(f)||_p = 0$.

Proof. Notice for some $g \in C_c(\mathbb{R}^n)$, we have

$$||\tau_{y+z} f - \tau_z f||_p \leq ||\tau_{y+z} f - \tau_{y+z} g||_p + ||\tau_{y+z} g - \tau_z g||_p + ||\tau_z f - \tau_z g||_p$$

and choose g_ϵ to be some function that $||g_\epsilon - f||_p < \epsilon$, so

$$\limsup_{y\to 0} ||\tau_{y+z} f - \tau_z f||_p \leq ||\tau_{y+z} f - \tau_{y+z} g||_p + \limsup_{y\to 0} ||\tau_{y+z} g - \tau_z g||_p + ||\tau_z f - \tau_z g||_p < 2\epsilon$$

and we are done. □

Definiton B.2.4 (Convolution)**.**

For f, g measurable on $\mathbb{R}^n$, the convolution $f * g$ is defined by

$$f * g(x) = \int f(x-y)g(y)dy$$

Proposition B.2.5 (Young's Inequality)**.**
If $f \in L^1$ and $g \in L^p (1 \leq p \leq \infty)$, then $f * g(x)$ exists for almost every x, $f * g \in L^p$ and $||f * g||_p \leq ||f||_1 ||g||_p$.

Proof. This can be checked easily using Minkowski's inequality for integrals. □

Proposition B.2.6. If p and q are conjugate exponents, $f \in L^p$ and $g \in L^q$, then $f * g(x)$ exists for every x, $f * g$ is bounded and uniformly continuous, and $\|f * g\|_u \leq \|f\|_p \|g\|_q$. If $1 < p < \infty$, then $f * g \in C_0(\mathbb{R}^n)$.

Proof. □

Definiton B.2.5 (Fourier Transform)**.**
For $f \in L^1(\mathbb{R}^n)$, define its Fourier transform by

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} \, dx.$$

Proposition B.2.7 (Basic properties of the Fourier transform)**.**
Let $f, g \in L^1(\mathbb{R}^n)$.

1. $\widehat{f * g} = \hat{f}\hat{g}$.
2. If $\tau_y f(x) = f(x-y)$, then $\widehat{\tau_y f}(\xi) = e^{-2\pi i y \cdot \xi} \hat{f}(\xi)$.
3. If $M_\eta f(x) = e^{2\pi i x \cdot \eta} f(x)$, then $\widehat{M_\eta f}(\xi) = \hat{f}(\xi - \eta)$.
4. If $f_a(x) = f(ax)$ with $a \neq 0$, then $\widehat{f_a}(\xi) = |a|^{-n} \hat{f}(\xi/a)$.

Proof. □

Proposition B.2.8 (Riemann–Lebesgue Lemma)**.** If $f \in L^1(\mathbb{R}^n)$, then $\hat{f} \in C_0(\mathbb{R}^n)$.

Proof. □

Proposition B.2.9 (Differentiation and multiplication)**.**
Suppose $f \in \mathcal{S}$.

1. $\widehat{\partial^\alpha f}(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$.
2. $\widehat{x^\alpha f}(\xi) = \left(\frac{i}{2\pi}\right)^{|\alpha|} \partial^\alpha \hat{f}(\xi)$.

Proof. □

Proposition B.2.10 (Inversion formula)**.** If $f \in \mathcal{S}$, then

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i x \cdot \xi} \, d\xi.$$

Proof. □

Theorem B.2.11 (Plancherel Theorem)**.** The Fourier transform extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$. In particular, for $f \in L^2(\mathbb{R}^n)$,

$$\left\|\hat{f}\right\|_2 = \|f\|_2 ,$$

and for $f, g \in L^2(\mathbb{R}^n)$,

$$\langle \hat{f}, \hat{g} \rangle_{L^2} = \langle f, g \rangle_{L^2}.$$

Proof. □

Theorem B.2.12 (Hausdorff–Young Inequality)**.** If $1 \leq p \leq 2$ and q is the conjugate exponent of p, then for $f \in L^p(\mathbb{R}^n)$,

$$\left\|\hat{f}\right\|_q \leq \|f\|_p .$$

Proof. □

Proposition B.2.13 (The Gaussian is its own Fourier transform)**.**

Let $f(x) = e^{-\pi|x|^2}$. Then

$$\hat{f}(\xi) = e^{-\pi|\xi|^2}.$$

Proof. □

Proposition B.2.14 (Fourier transform on the Schwartz space)**.** The Fourier transform maps $\mathcal{S}$ bijectively onto itself.

Proof. □

Proposition B.2.15 (Approximate identity via Gaussians)**.** Let $G_t(x) = t^{-n/2} e^{-\pi|x|^2/t}$ for $t > 0$. Then for $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$,

$$G_t * f \to f \quad \text{in } L^p(\mathbb{R}^n) \text{ as } t \to 0.$$

Moreover,

$$\widehat{G_t}(\xi) = e^{-\pi t|\xi|^2}.$$

Proof. □

References